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Emerging Trends in Snow Avalanche Modeling

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Abstract

Spatial variability of terrain, snow pack, and weather parameters play a critical role in the avalanche release process. Various techniques have been attempted world over to model the process. In the present study, the authors have discussed in detail a method which could integrate the static terrain-related parameters, and dynamic snow-meteorological parameters within a single modeling framework. This study adopts Analytical Hierarchy Process (AHP) and Ideal Point Analysis (IPA) techniques for avalanche hazard zonation, identification of avalanche triggering/formation zone and delineation of avalanche path with static terrain data, and risk along the road and settlements with additional snow-meteorological data. A case study has been presented for a remote snow-bound avalanche prone highway in Himalayan region.

Keywords: Spatial Modeling, Avalanche hazard zonation, Risk assessment, Multicriteria Decision-Making methods.

1.0 INTRODUCTION

1.1 Avalanche: Phenomenon

An avalanche is a massive slide of snow, ice, rock or debris down a mountainside. It can be provoked by an earth tremor, extreme precipitation or man-made disturbances (such as a mountain skier). The impact of the falling material and the frontal pressure produced by the flow can cause extensive damage to anything in its path. In the case of a snow avalanche, the new snow that accumulates on top of another heavy layer of snow can begin to slide down the mountainside. The risk of an avalanche can be reduced by building a snow shed — a barrier made of rocks, soil and other materials — or by triggering a

controlled avalanche at a time when no one is on the mountain. Minimal requirements for the occurrence of an avalanche are snow and an inclined surface, usually a mountainside. Most avalanches occur on slopes between 30 and 45°.

Avalanches are a well-known geophysical threat in snow-bound mountainous areas. Spatial variability of terrain, snow pack, and weather parameters play a critical role in the avalanche release process. This complex process has been studied in depth by researchers worldwide. Ganju et. al. (2002) studied in detail the characteristics of Himalayan avalanches. Majority of avalanche releases that killed people took place from slopes between 30 to 45 degrees. Of these, the slopes of 31-35 deg claimed the maximum (33%) deaths. Avalanche accidents take place from all types of aspects in the Western Himalayan region. However, northerly, southeasterly and southwesterly slopes have taken the maximum toll constituting about 50% of the total accidents recorded so far.

1.2 Avalanche: Modelling

Various techniques have been adopted to model the process. Some of the techniques enable one to analyze the problem in spatial domain addressing: Where an avalanche triggering zone will be formed, Which path the avalanche would take, What is the risk levels of a road or settlement, How much would be the damage; whereas some techniques enable one to analyze the problem in time domain addressing: When an avalanche would probably happen, Is today an avalanche day or not? While there is a merit in the two basic approaches, it is quite evident that the problem has to be addressed both in spatial and time domain together.

While modeling techniques like Fuzzy logic, Artificial Neural Networks, Expert Systems, Nearest-neighbor methods are in vogue in time-domain, each one of them has its own merits and demerits, for example: Artificial Neural Network techniques though could effectively capture the non-linear relationships, it uses only quantitative snow-met parameters; with rule-based expert systems it is difficult to state all possible combination of rules due to number of parameters involved in the avalanche release process process (Naresh & Pant, 1999). Other statistical methods like Nearest Neighbor method (Buser, 1983; Purves et al., 2003) are based on historical data and so far have been found to be most successful for operational use, but still do not offer the complete solution.

Multi criteria decision-making (MCDM) problems such as the prediction of time and place of occurrence of avalanches, involve a set of alternatives that are evaluated on the basis of a set of evaluation criteria. The objective of using MCDM techniques is to help find solutions to decision problems characterized by

multiple-choice alternatives, which can be evaluated by means of performance characteristics called decision criteria. Because much of this modeling approach deals with evaluating location choice alternatives on the basis of suitability criteria, considerable attention in the last decades was devoted to integrating MCDM with GIS software. These efforts resulted in a number of prototypes and a few commercial software packages like IDRISI and ExpertChoice.

1.3 About the present study

Alternate approaches to GIS-based multi criteria analysis have been suggested to overcome the problem of weighting and data integration. Combining different factors, some exclusionary and some expedient, requires a weighting factor. Analytic Hierarchy Process (AHP) is an approach that can be used to determine the relative importance of a set of activities or criteria (Saaty 1990). AHP has been widely used in the multi-criteria decision-making process in varied fields (Saaty and Vargas, 1990). This study is an attempt to adopt Analytical Hierarchy Process and Ideal Point Analysis techniques for

- a) Avalanche Hazard Zonation with static terrain data
- b) Identification of avalanche triggering / formation zone
- c) Delineation of avalanche path
- d) Risk along the road, settlements
- e) Avalanche Danger Assessment with terrain, snow-met data

This attempt thus integrates the static terrain-related parameters, and dynamic snow-meteorological parameters within a single modeling framework. The method was employed over a remote snow bound avalanche prone highway section in north-western part of Indian Himalaya. The overall results obtained are presented in subsequent sections.

2.0 MULTI-CRITERIA DECISION-MAKING TECHNIQUES

Multicriteria decision-making (MCDM) techniques are found to be suitable to handle avalanche hazard zonation problems. Multiple criteria which would influence avalanche hazard zonation would include terrain-related parameters like slope, aspect, morphology, land use/cover etc. Spatial modeling involves integrating various parameters by assigning weightages to the parameters and for classes within each parameter. Analytic Hierarchy Process (AHP) has been identified as a weighting strategy. Different domain experts give their opinion or experts can come to a consensus and give their opinion. Compromise Programming (CP) technique has been identified for data integration (Novaline et

al. 2006, Deekshatulu et al. 1999). The resultant composite layer obtained after integration provides information on the avalanche hazard and is grouped into 6 hazard severity classes.

2.1 Analytic Hierarchy Process (AHP)

AHP is an approach that can be used to determine the relative importance of a set of activities or criteria.

The first step of the AHP is to form a hierarchy of objectives, criteria and all other elements involved in the problem. Once the hierarchical structure has been formed, comparison matrices are developed. These are evaluations made by the decision-makers on the intensity of difference in importance, expressed as a rank number on a given numerical scale, for each level in the hierarchy. From these weights, priorities are determined. An expert is asked to make pairwise comparisons between two factors at a time, decide which factor is more important, then specify the degree of importance on a scale between 1 and 9 in which 9 is most important. These evaluations result in reciprocal matrices of the components of each level against the items in the level above. Consistency of the matrix is checked and Eigen values of the matrix are found. Upon normalization of the Eigen vector corresponding to maximum Eigen value, each factor coverage has only one weight associated with it.

2.2 Ideal Point Analysis / Compromise Programming

An important problem in spatial modeling is how to efficiently integrate data from various sources. Weighted linear additive model is the one that is widely used for data integration and is done with the help of algebraic functions available in any commercial GIS package. Here, a total compensation between criteria is assumed, meaning that a decrease of one unit on one criterion can be totally compensated by an equivalent gain on any other criteria. Compromise Programming technique, is a method to arrive at non-compensatory solution. It measures the deviations from the ideal point in each data layer and a min-max rule is applied wherein minimum of the maximum weighted deviations are sought for getting a composite layer. The best compromise solution is defined as that which is at the minimum distance from the theoretical ideal.

3.0 STUDY AREA

The proposed methodology was applied for avalanche hazard zonation and other outputs over Chowkibal-Tangdhar axis (CT axis) in Indian Western Himalaya (Figure 1). This 35 Kms long highway stretch is highly avalanche prone during winter months and is affected by 26 major avalanche sites. There are three

snow-meteorological observatories situated in the area from where necessary data was obtained for this study.

4.0 METHODOLOGY AND RESULTS

4.1 Derivation of Terrain Parameters

Terrain-related parameters which includes, Slope, Curvature, Ruggedness, Aspect, Altitude, Landcover, Vegetation density were automatically derived. Ruggedness of the terrain was computed by the Compounded Terrain Complexity Index which computes the topographic turbulence in a 3X3 window of elevation nodes, considering range in elevation, variance in heights, curvature, and rugosity (ratio of the surface area to the projected area). Land cover classes, viz., barren land, grass and trees classes were derived from the IRS-P6 LISS-4 data using supervised maximum likelihood classification technique. And then the vegetation density was derived by using this trees classes derived from LISS-4 data. A moving window of 5*5 size was selected for estimating the vegetation density.

4.2 Avalanche Hazard Zonation Map

A static terrain-based avalanche hazard zonation map was generated using terrain-related parameters which includes, Slope, Curvature, Ruggedness, Aspect, Altitude, Landcover, Vegetation density. Each parameter has sub-categories. Figure1 shows the AHP tree formulated for this problem. The various levels of the tree are as under -

Level 1 – Objective

Level 2 – Experts who can give the opinions of parameters and categories within parameters

Level 3 – Parameters

Level 4 - Interdependency among the parameters can also be studied

Level 5 - Categories within each parameter.

Experts' opinion on the importance of the parameters for avalanche hazard was obtained through AHP and weights were derived. The data layers were integrated with Ideal Point Analysis and Avalanche Hazard Zonation map showing six hazard classes was obtained (Figure 3).

4.3 Characterization of Hazard Polygons within the Basin

Basin-wise avalanche hazard characterization was done by delineating basins using Digital Elevation Model. First three best Avalanche Formation Zone or

Release areas were combined and hazard polygons were derived as shown in the Figure 4. There are 9 hazard polygons and depending on the sun-facing aspect, and snow-met parameters a subset of them may avalanche at a particular instant of time. Avalanche Path has been derived for each hazard polygon. As seen in the Figure 5, showing the perspective view of the terrain, each polygon falls in different aspect and might be triggered at the same time or at different instants.

Hazard level at formation zone is determined by considering the frequency of avalanche, settlement proximity, roads in the formation zone, distance of formation zone from road and visibility of formation zone from road.

4.4 Risk to Road

The risk to pedestrians and vehicular traffic on roads in the avalanche prone areas has been estimated. Factors considered for modeling risk to road include: frequency of avalanche, area of hazard polygons contributing to road basin-wise, avalanche path intersecting the road, gradient of road, stretch of hazard polygon close to road, and stretch of high risk zone in the path. Risk along road is categorized into five risk classes (Figure 6).

4.5 Avalanche Danger Assessment with terrain and snow-meteorological data

Avalanche danger assessment is carried out on daily basis during winter months when avalanche slopes are loaded with snow. It involves the analysis of snow, met and terrain parameters. Depending upon their threshold values, different parameters are categorized as critical, transitional or non-critical from avalanche activity point of view. These threshold values are arrived at by analyzing the past data of these parameters for each station/area. Following parameters are taken as input parameters for the assessment of avalanche danger: Cumulative Fresh snow of last 24hrs, Storm Snow, Standing snow, Snow settlement during last 24 hrs, Snow fall intensity, Snow surface temperature, Free penetration, Average wind speed, and Radiation. Hazard level at formation is combined with avalanche danger assessment obtained with snow-meteorological data and an overall assessment of danger is obtained (Figure 7).

5.0 CONCLUSIONS

Traditionally GIS, means a system capable of storing, manipulating, analyzing and displaying spatial data. It lacks the ability to model a dynamic phenomenon in spatial-temporal domain. But it can act as a platform on which further modeling

capabilities can be built. In the present study, an attempt has been made to enhance the spatial modeling capability of a GIS to address spatial dynamic modeling problem, through integration of Multi-Criteria Evaluation procedures.

The paper presents a detailed account of the overall integrated GIS-MCDM approach to enhance the spatial modeling capability of GIS. Integrated GIS-MCDM approaches can enhance the current relatively poor spatial modeling capability of GIS. MCDM models can be completely developed within GIS for easily accessing the information stored in the GIS database during the modeling processes. Constraints for modeling can be defined using GIS with other collateral data. Therefore, the development of MCDM within GIS greatly enhances the ability of spatial modeling within GIS. Through the present case study, the integrated GIS-MCDM framework proves to be a promising environment, wherein a variety of spatial-dynamic phenomena can be modeled.

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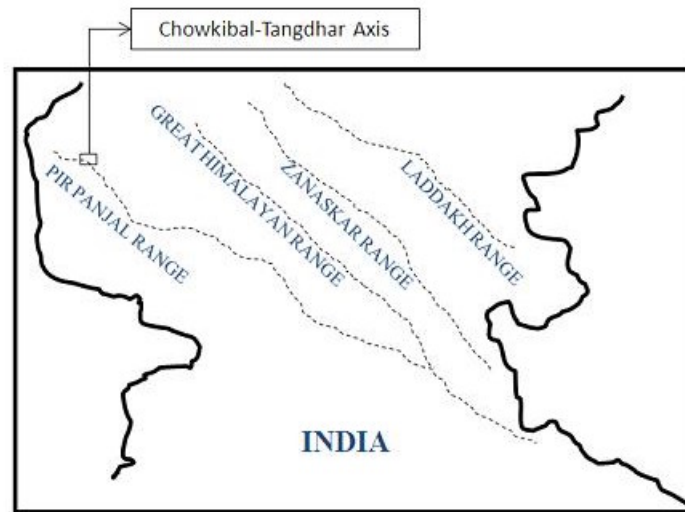


Figure 1: A perspective view of study areas - Chowkibal-Tangdhar axis

Knowledge Engineering adopting AHP

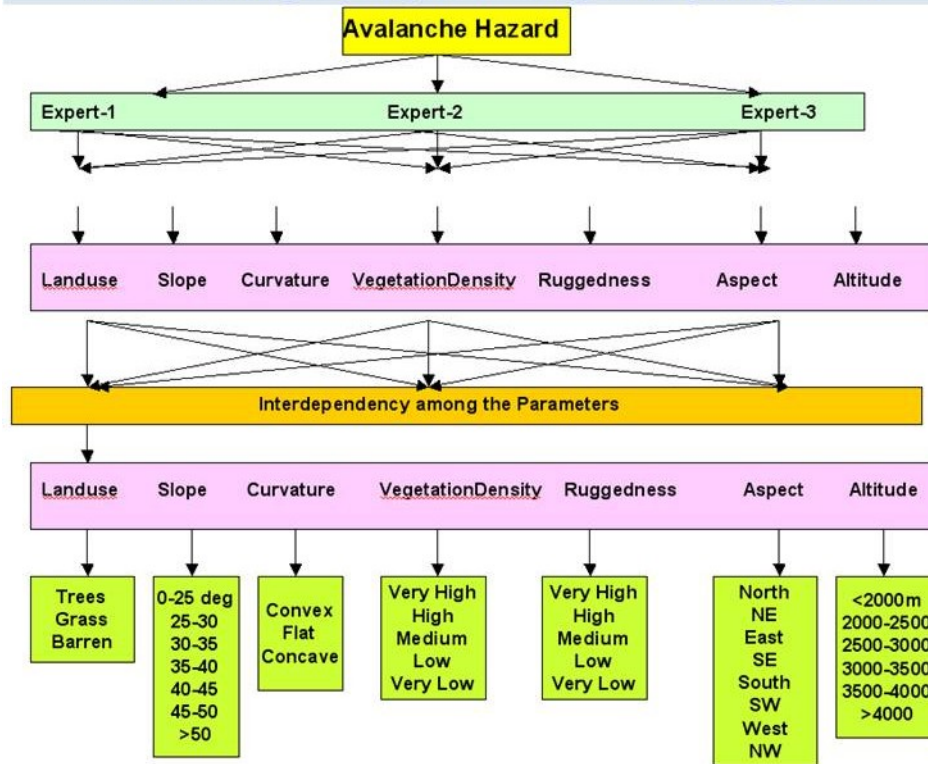


Figure 2. AHP Tree

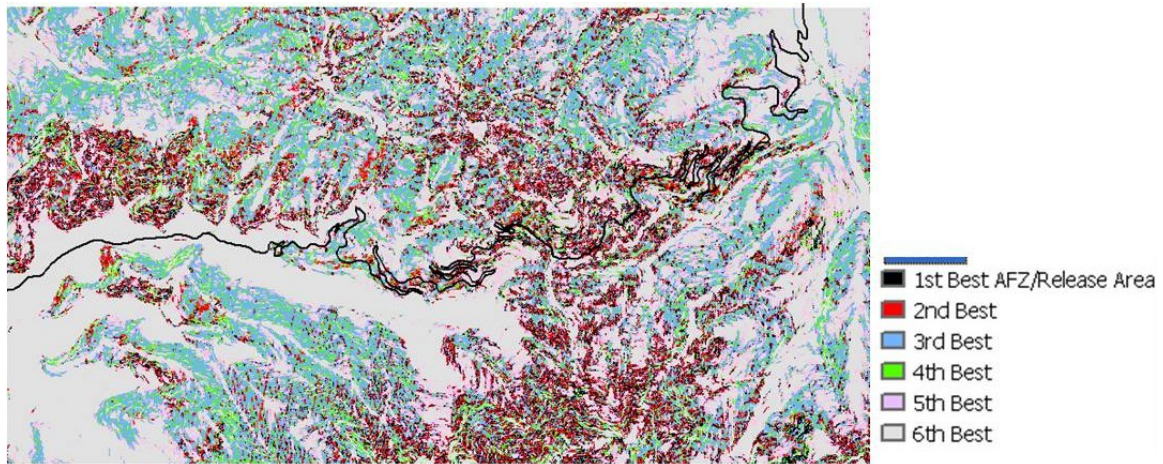


Figure 3. Avalanche Hazard Zonation Map

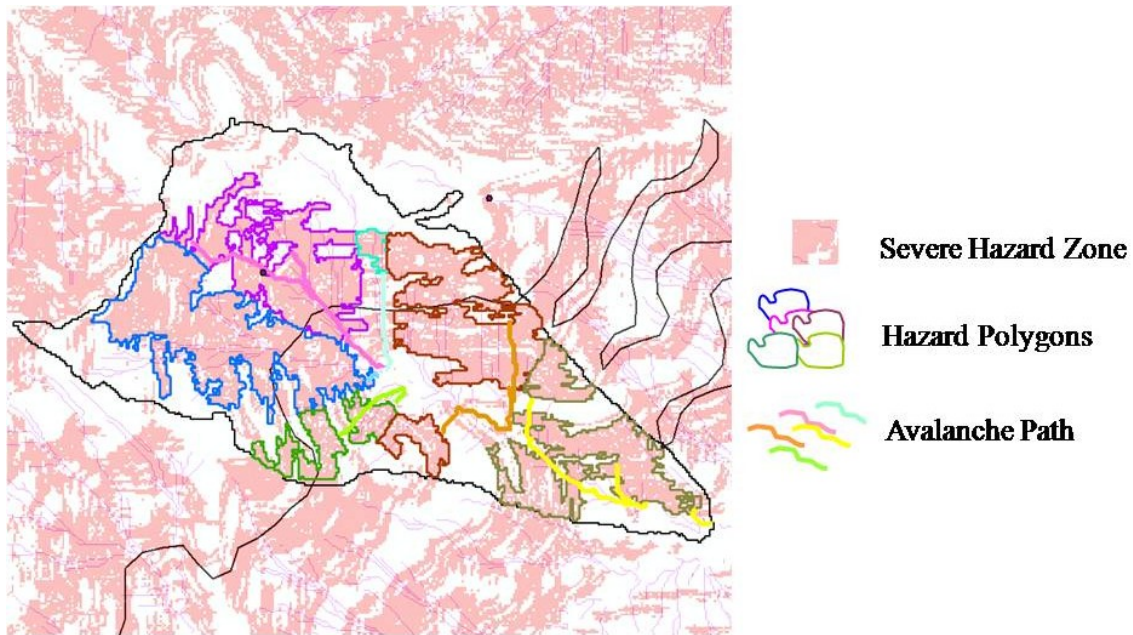


Figure 4. Basin-wise Characterization of Hazard Polygons

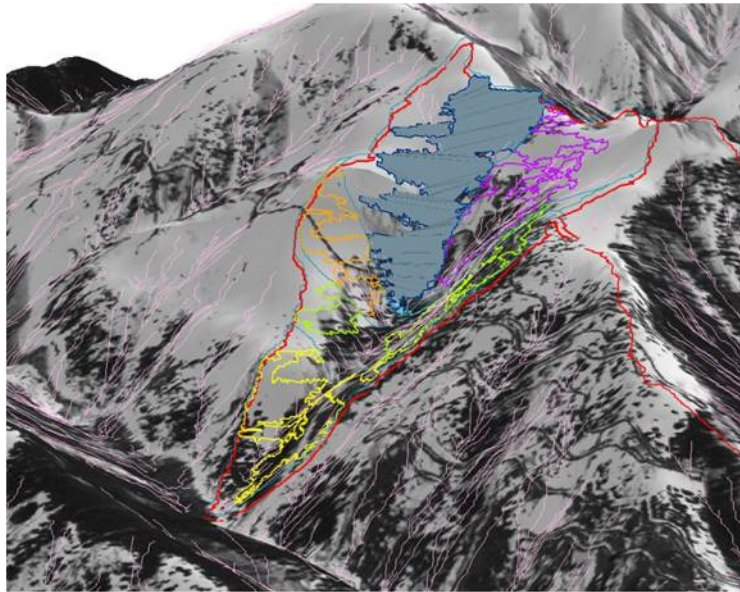


Figure 5. Perspective view of the basin and hazard polygons

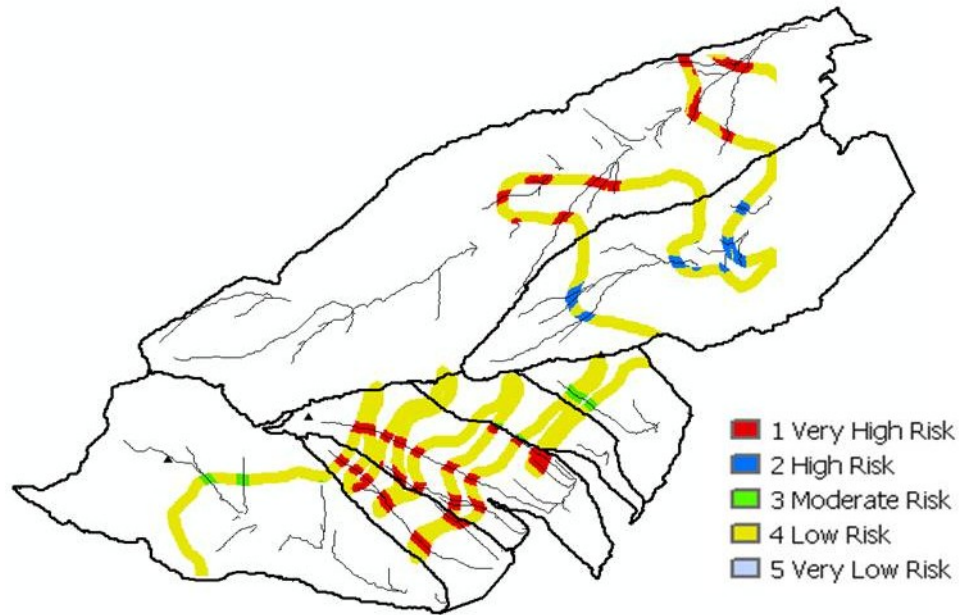


Figure 6. Risk along road

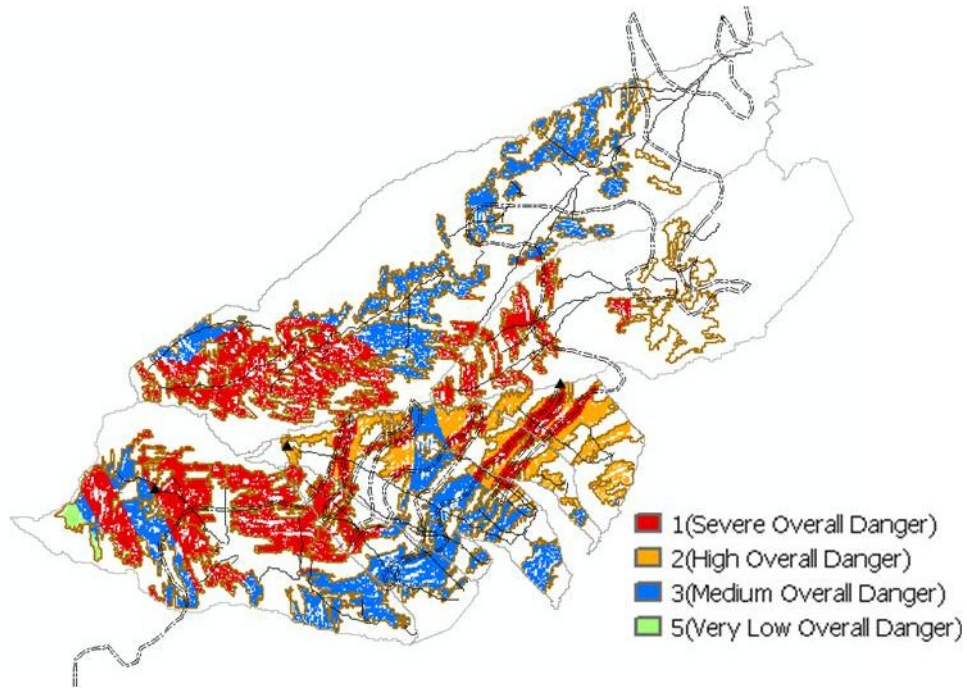


Figure 7. Overall Danger Level at Formation Zone Polygons



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